Digital Parking Control System: A Design & Implementation

EEE 3101: Digital Logic Circuits

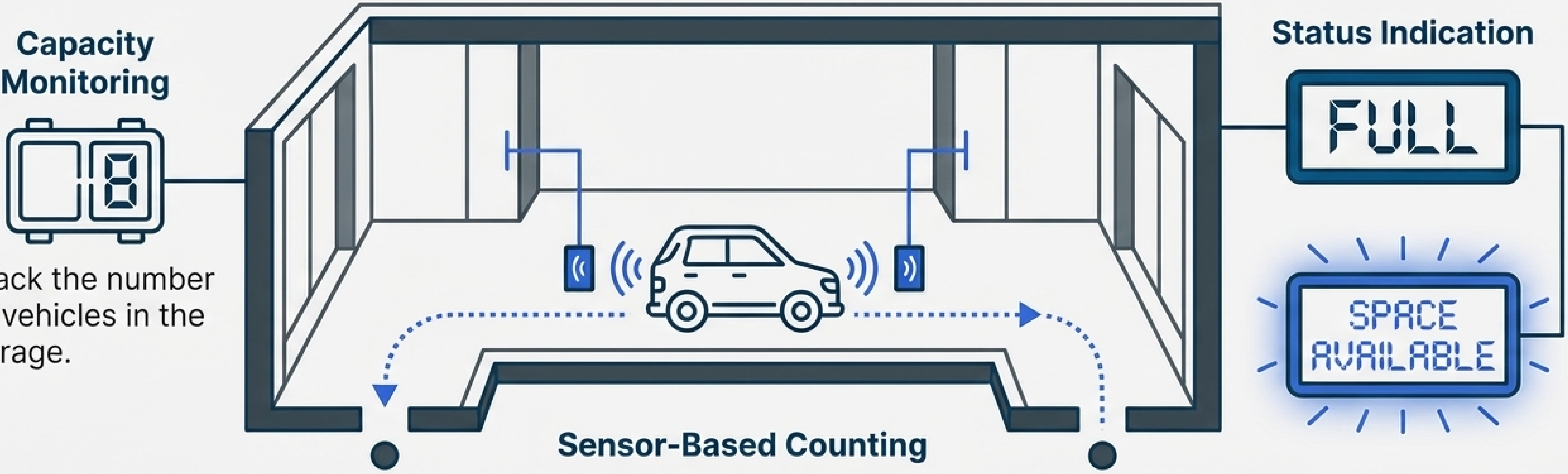
Shoriful Islam
Student ID: 23-50357-1

The Engineering Challenge: Automating Parking Management

Mission: Design a digital system to monitor and control a parking garage with a finite, pre-determined capacity.

Capacity Monitoring

Track the number of vehicles in the garage.



Status Indication

Sensor-Based Counting

Use entry and exit sensors to update the vehicle count.

NotebookLM

System Parameters Derived from Student ID: 23-50357-1

All system variables are calculated based on the digits of the student ID `ab-cdefg-h` (23-50357-1).

Maximum Vehicle Capacity

`b + f + h = 3 + 5 + 1`

`b + f + h = 3 + 5 + 1`

9 Cars

Requires a 4-bit counter ($2^4 = 16 > 9$).

Blinking Display Frequency (f)

`a + c + 5 = 2 + 5 + 5`

`a + c + 5 = 2 + 5 + 5`

12 Hz

Blinking Display Duty Cycle (D)

`P = (a+b+d+e+g)*2 = (2+3+0+3+7)*2 = 30`

`D = 100 - P = 100 - 30`

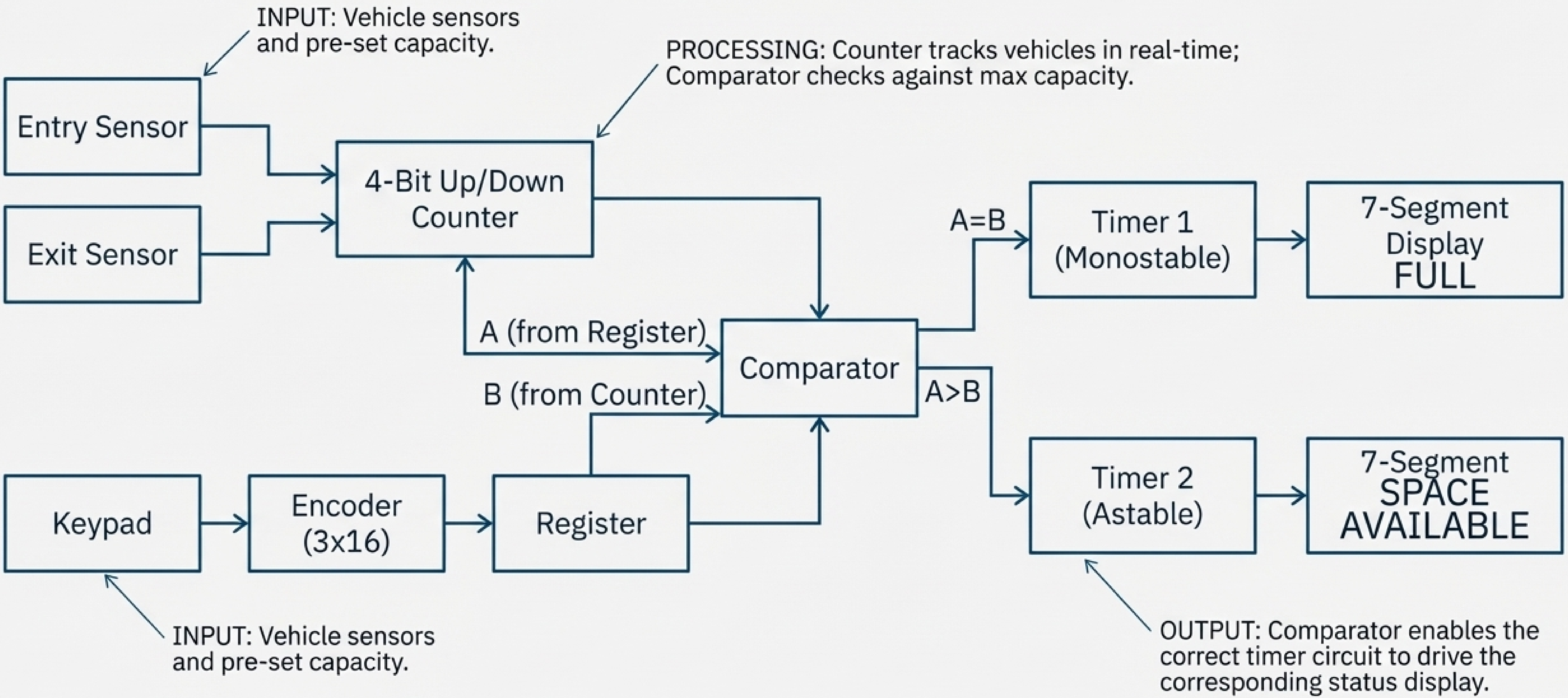
70 %

Calculated Blink Timing

On-Time (TH) = **58.8 ms**

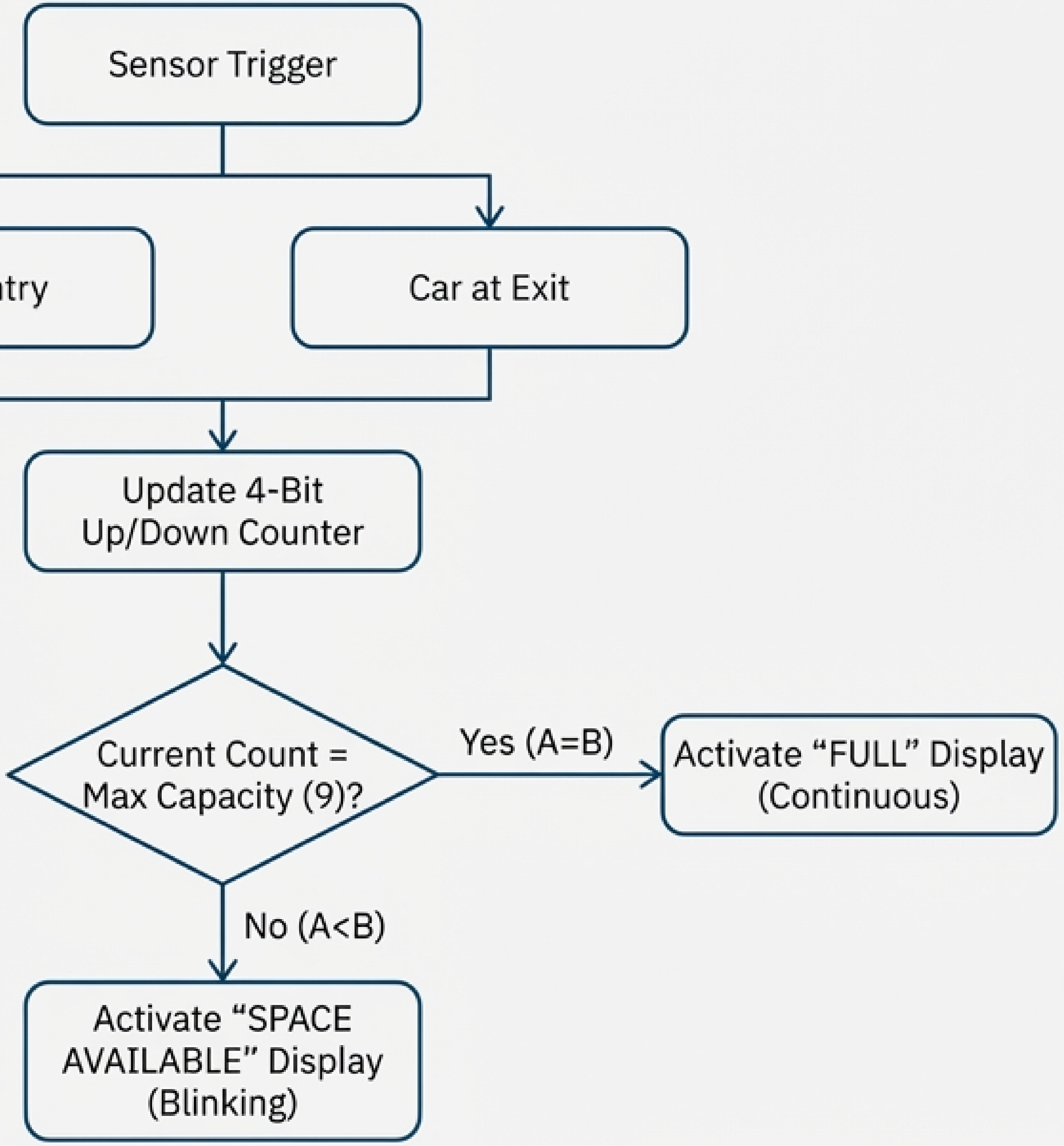
Off-Time (TL) = **25.2 ms**

The System Architecture



NotebookLM

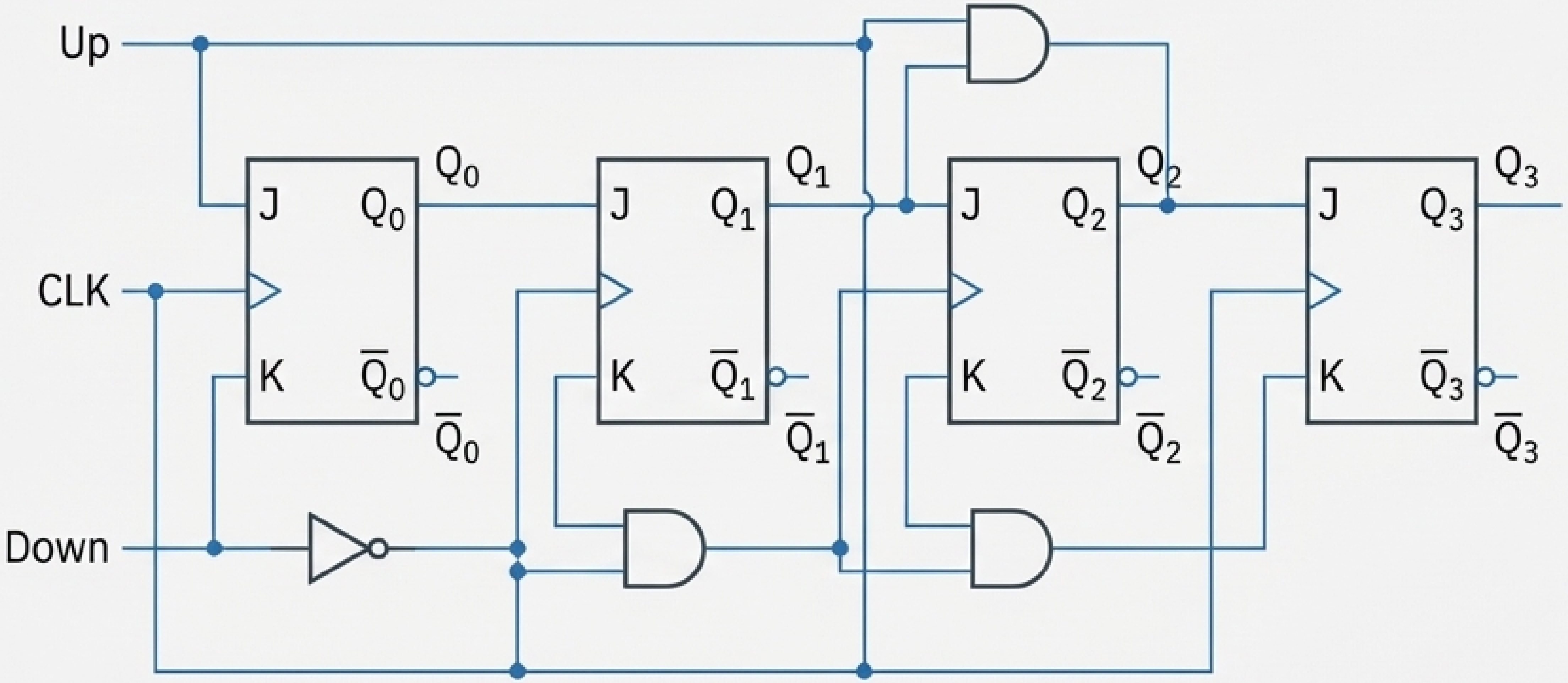
System Operational Logic



NotebookLM

The Core Component: 4-Bit Synchronous Up/Down Counter

A custom counter designed with J-K Flip-Flops to track vehicle count.

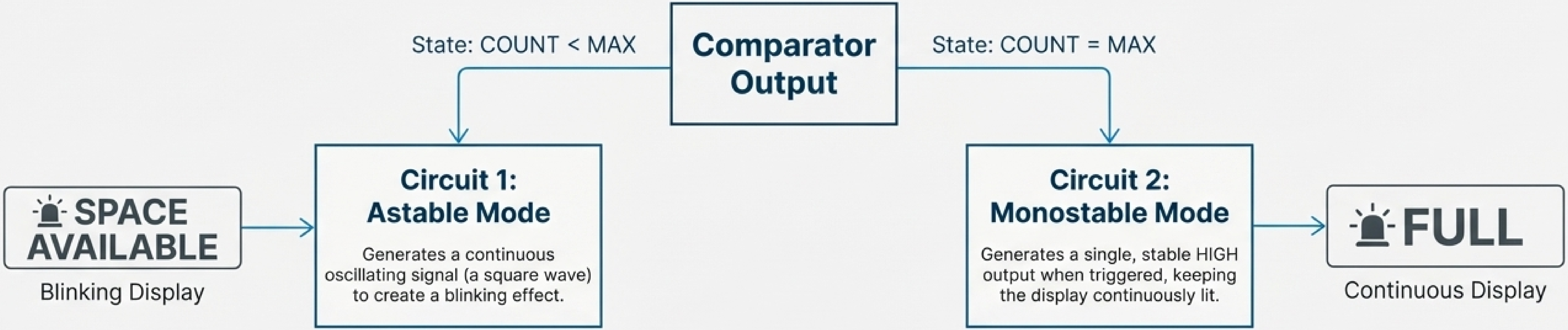


Functional Description

- **Function:** Increments count when a car enters (**Up** signal is HIGH) and decrements when a car exits (**Down** signal is HIGH).
- **Synchronous Design:** All flip-flops change state simultaneously on the same clock edge, preventing ripple errors.
- **Capacity:** As a 4-bit counter, it can represent states from 0 (binary 0000) to 15 (binary 1111), easily accommodating the required maximum capacity of 9.

Display Control Using 555 Timers

Strategy: Two distinct 555 timer circuits are used to manage the two different display states, controlled by the output of the system's comparator.



Calculating Components for the Blinking Display

Design Goals: Frequency (f) = 12 Hz | Duty Cycle (D) = 70%
Chosen Component: Capacitor (C) = 250 μF

1. Time Period (T): $T = \frac{1}{f} = \frac{1}{12\text{ Hz}} = \mathbf{83.3\text{ ms}}$

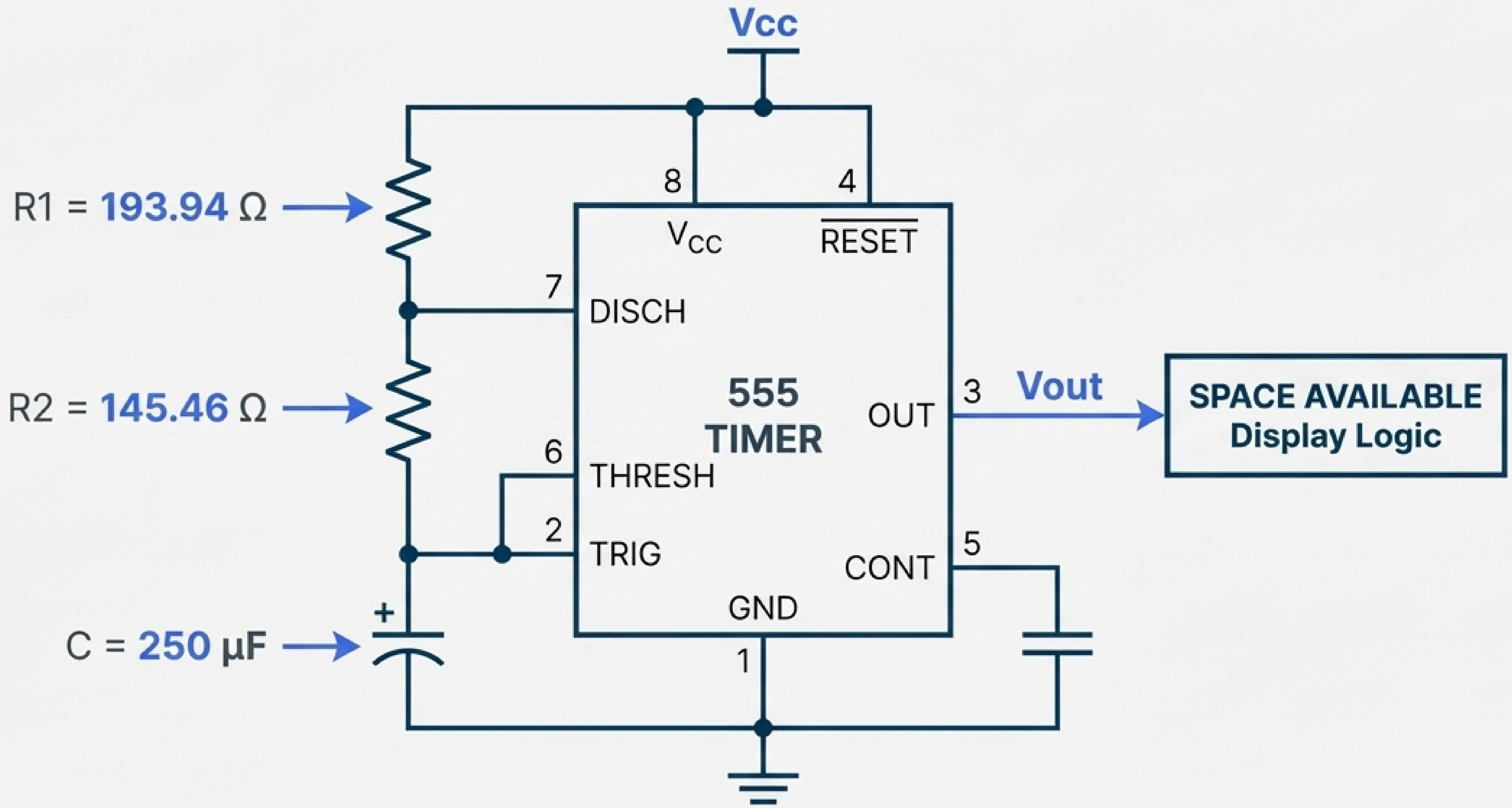
2. On-Time (T_H): $T_H = D \times T = 0.70 \times 83.3\text{ ms} = \mathbf{58.3\text{ ms}}$

3. Off-Time (T_L): $T_L = T - T_H = 83.3\text{ ms} - 58.3\text{ ms} = \mathbf{25\text{ ms}}$

4. Resistor R2 Calculation: $R2 = \frac{T_L}{0.693 \times C} = \frac{25.2\text{ms}}{0.693 \times 250\mu\text{F}} = \mathbf{145.46\ \Omega}$

5. Resistor R1 Calculation: $R1 = \frac{T_H}{0.693 \times C} - R2 = \frac{58.8\text{ms}}{0.693 \times 250\mu\text{F}} - 145.46\ \Omega = \mathbf{193.94\ \Omega}$

Timer 1: Astable Circuit for “SPACE AVAILABLE”



NotebookLM

Programming System Behavior with a Finite State Machine

Clock Period: 1 ms
Requirement: Generate output A for T_H and output B for T_L.
Timing Constraint: For this FSM task, T_H and T_L are capped at **4 ms**.

State Table

Current State			Output B		
Reset	**Current State**	**Input**	**Next State**	**Output A**	**Output B**
1	-	-	P0	1	0
0	P0	0-3	P0	1	0
0	P0	4	P1	0	1
0	P1	0-3	P1	0	1
0	P1	4	P0	1	0

State Diagram

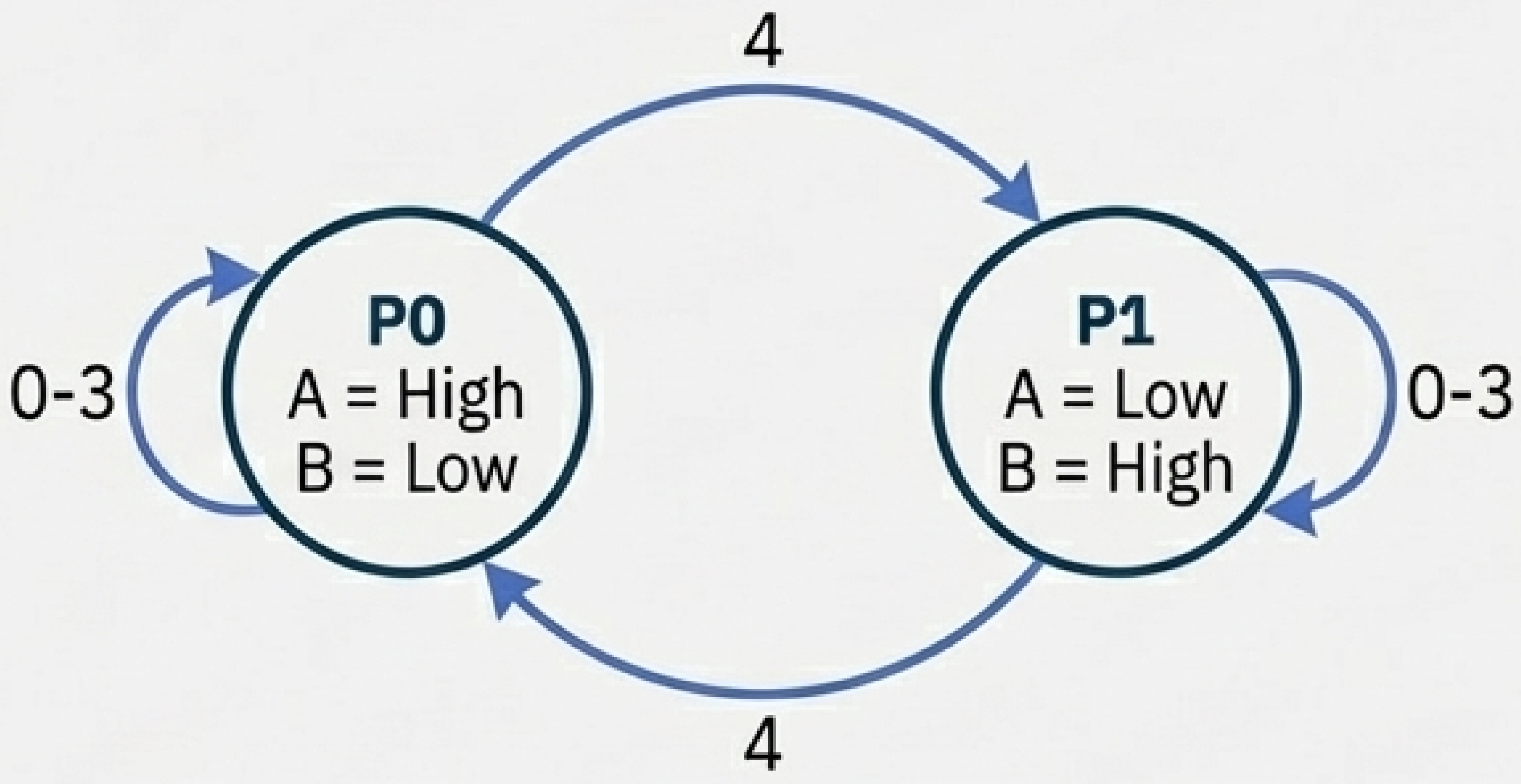


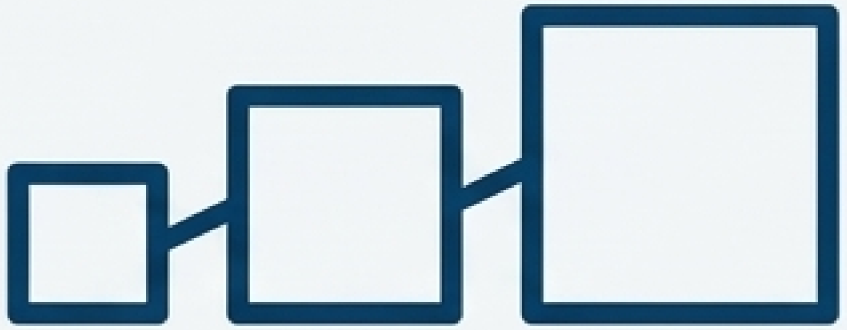
Fig: State Diagram for FSM.

Implementation in Hardware Description Language (HDL)

Module Name: parking_controller. **Inputs:** clk, reset, entry, exit. **Outputs:** Full, SPACE_AV.
Function: The code defines the state machine logic to update the vehicle count based on sensor inputs and control the Full and SPACE_AV output signals.

```verilog
// Verilog HDL Code:
// Assign 4-bit Counter using Flip-Flops to store
// current vehicle count.

module parking_controller(
    input wire clk, reset, entry, exit,
    output wire Full, SPACE_AV
);

    reg [1:0] P_state;
    reg [1:0] N_state;

    assign N_state[1] = ~reset & (~P_state[1] & entry) | (P_state[1] & ~exit);

    always @ (posedge clk) begin
        P_state[0] <= N_state[0];
        P_state[1] <= N_state[1];
    end

    assign Full = P_state[1] & P_state[0];
    assign SPACE_AV = ~P_state[1];

endmodule
```

A Critical Review: Limitations of the Design

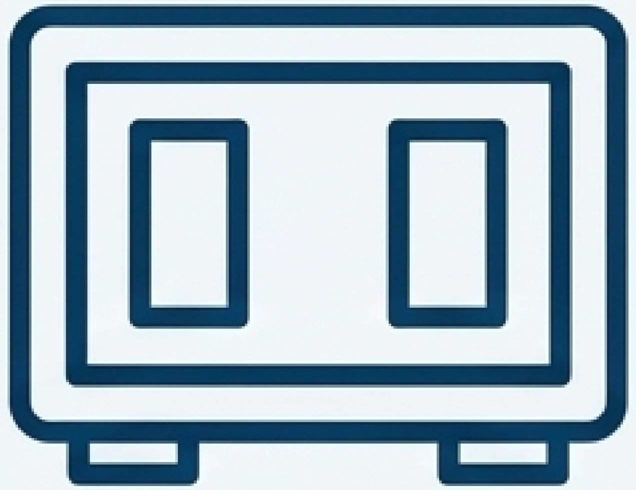


Scalability

The design, based on discrete flip-flops and counters, becomes overly complex and difficult to manage as the required parking capacity increases significantly.

Reliability

The system has single points of failure. The malfunction of a critical component, such as the main counter IC or a 555 timer, could disable the entire system or lead to incorrect status displays.



Information Granularity

The display is binary, only indicating 'FULL' or 'SPACE AVAILABLE.' It does not provide more useful information to drivers, such as the exact number of free spaces.

Envisioning a More Robust System: Enhancements & Impact

Proposed Improvements

- Replace discrete logic with a microcontroller (e.g., Arduino) for superior scalability and flexibility.
- Implement a numerical display to show the exact count of available spots.
- Install sensors in each individual parking bay for precise, real-time occupancy data.
- Integrate Wi-Fi or Bluetooth to enable a mobile app for remote availability checks.

Impact of Specification Change

Impact Area	Analysis
Complexity & Cost	Requires more components and sophisticated software, leading to a higher initial system cost.
User Experience	Delivers far more valuable and actionable information to drivers, reducing search time.
Maintainability	Software on a microcontroller is significantly easier to debug, update, and modify than hardware logic is to rewire.

Project Summary & Accomplishments

- **Successfully designed** a complete digital parking control system tailored to specific, calculated constraints derived from a unique student ID.
- **Demonstrated proficiency** in key areas of digital design, including sequential logic (counters, FSMs), combinational logic (comparators), and analog interface circuits (555 timers).
- **Architected** a full system solution, progressing from a high-level block diagram through detailed circuit schematics to final implementation in Hardware Description Language.
- **Critically evaluated** the completed design, identifying its inherent limitations and proposing clear, actionable pathways for future enhancement and scalability.

Thank You

Questions & Discussion

Shoriful Islam
Student ID: 23-50357-1
shoriful.islam.cse@aiub.edu